

ANIMAL CLASSIFICATION USING K-NEAREST NEIGHBOURS (KNN)

1. OBJECTIVE

The objective of this assignment is to implement and evaluate the K-Nearest Neighbours (KNN) algorithm to classify animals into different types based on biological characteristics such as hair, milk, eggs, feathers, legs, and others using the Zoo dataset.

2. EXPLORATORY DATA ANALYSIS (EDA)

The dataset consists of 101 animals with 17 features. A count plot was used to examine class distribution, showing that Mammals (Type 1) are the most frequent class. A correlation heatmap revealed strong relationships between features such as 'milk' and 'hair'.

3. OUTLIER ANALYSIS

Outlier detection was performed using boxplots and the IQR (Interquartile Range) method on the numerical feature 'legs'. While values such as 5 or 8 legs were detected outside the typical range, these represent natural biological variations rather than data errors. Therefore, no outliers were removed.

4. DATA PREPROCESSING

The 'animal name' column was removed as it does not contribute to prediction. Missing value checks confirmed that no null values were present. Since KNN is distance-based, StandardScaler was applied to normalize features so that all variables contribute equally to distance calculations.

5. MODEL IMPLEMENTATION

The dataset was split into 80% training and 20% testing sets. The KNeighborsClassifier from scikit-learn was implemented using Euclidean distance. The optimal K value was selected using the Elbow Method, and K=5 provided the best balance between bias and variance.

6. PERFORMANCE EVALUATION

The model achieved approximately 95% accuracy on the test dataset. Precision, Recall, and F1-score values were high across most classes, indicating strong classification performance. A confusion matrix confirmed that misclassifications were minimal.

7. DECISION BOUNDARY VISUALIZATION

Since the dataset contains multiple features, Principal Component Analysis (PCA) was used to reduce dimensionality to two components for visualization purposes. The decision boundary plot clearly shows how KNN groups similar animals together based on nearest neighbors in the transformed feature space.

8. STRENGTHS AND WEAKNESSES OF KNN

Strengths: - Simple and easy to implement - No training phase (lazy learner) - Works well for smaller structured datasets Weaknesses: - Computationally expensive for large datasets - Sensitive to outliers - Requires feature scaling

9. CONCLUSION

The KNN algorithm performed effectively on the Zoo dataset due to the distinct binary nature of most features. Proper scaling, outlier analysis, and selection of an optimal K value ensured strong model performance. Decision boundary visualization further validated the classifier's behavior.